Fraud Analysis

By Robert Norris

Data source

Data Foundation

Source: IEEE-CIS Fraud Detection (Vesta Corporation) via Kaggle.

- Scale: Analyzed a massive corpus of 506,691 real-world e-commerce transactions.
- Complexity: Evaluated over 390 features including device identity, network metadata, and transaction signatures.
- Focus: Specialized in the intersection of identity anonymity (email domains) and hardware spoofing (mobile/desktop builds).

Goals

Analyze and Identify the riskiest email domains, devices and cards and why they are so risky

Emails

Despite having the second lowest total attempts(number besides the bar) in the graph, protonmail has the highest fraud rate at 95%, why is this?

Identity Obfuscation: End-to-end encryption and "No-Logs" policies allow fraudsters to operate with total anonymity.

Low Barrier to Entry: Ability to create accounts without linked Phone/SMS verification enables "Mass Account Creation" for bot attacks.

Regulatory Shield: Swiss privacy protections make cross-border fraud investigations significantly more difficult for retailers and banks.

Comparative Risk Analysis: Email Domain Fraud Velocity vs. Volume

Devices: Mobile vs Desktop

Compared to Desktops, mobile devices had a higher average fraud rate, why is this?

Mobile devices while not inherently “weaker” allow attackers to hide behind standardized device fingerprints and rotating cellular IP addresses, making individual identification much more difficult than on Desktop.

Mobile devices are also easier to access and acquire when compared to desktops

Comparative Risk Analysis: Mobile Fraud

Device Type	Device Info	
mobile	Z835 Build/NMF26V	100.025
	XT890 Build/9.8.2I-50_SML-25	100.08
	VS5012 Build/NRD90M	100.034
	TR10CS1 Build/JDQ39	100.010
	SM-N920A Build/MMB29K	100.021
	SM-J700T Build/MMB29K	100.08
	RCT6203W46 Build/KOT49H	100.09
	NOKIA	100.010
	Nexus 6 Build/MOB30M	100.011
	G3123 Build/40.0.A.6.175	100.012
	Blade L3 Build/KOT49H	100.012

Comparative Risk Analysis: Desktop Device

Device Type	Device Info	
desktop	CHC-U03 Build/HuaweiCHC-U03	100.0028
	Ilium Pad T7X Build/LMY47I	100.004
	Lenovo TB3-710I Build/LMY47I	100.002
	SM-A300M Build/KTU84P	100.003
	SM-J500M Build/LMY48B	100.007
	Z813 Build/LMY47O	100.009
	LG-D320 Build/KOT49I.V10a	96.8361
	SM-A300H Build/LRX22G	90.9120
	LG-H320 Build/LRX21Y	85.716
	SM-G900M Build/KOT49H	83.335

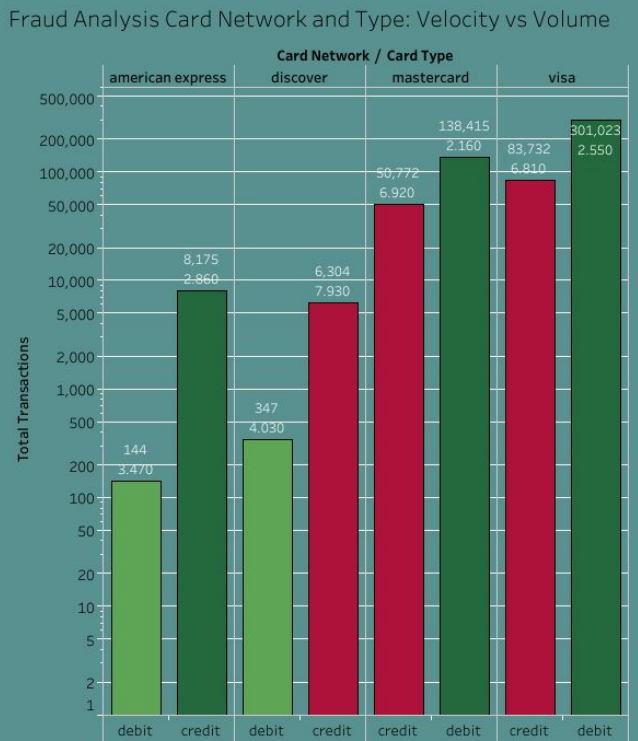
Cards: Network and Type

Despite credit cards having higher overall attempts(top number), credit cards had higher average fraud rates(bottom number), why is that?

Institutional vs. Personal Risk: Credit cards draw from bank-issued lines of credit, creating a layer of separation that delays detection compared to direct-draw Debit accounts.

Extended Detection Window: Consumer protections (Zero Liability) often lead to slower reporting of stolen credentials, giving attackers more time to exploit the card.

Liquidity Advantage: Fraudsters prefer Credit for higher transaction limits and the ability to "wash" funds through multiple purchases before the billing cycle closes.



Overall Risk score

Developed a custom Risk Score by calculating the mean fraud rate across three dimensions: Email Domain, Device Profile, and Payment Type.

The risk score found protonmail had the highest overall fraud rate(top number) and highest risk score(bottom number), regardless of the payment type or device used

Comparative Risk Analysis: Email, Cards and Devices

			R emaildomain																	
Device Type	Card Type	Card Network	protonm..	mail.com	outlook.c..	icloud.co..	outlook.es	gmail.com	live.com...	hotmail...	yahoo.co..	yahoo.es	yahoo.fr	earthlink..	mac.com	live.com	anonymo..	ymail.com	aol.com	bellsout..	me.com
desktop	credit	discover			15.386			7.043	100.008	6.333	21.746				40.006					20.006	14.296
		visa	92.318	12.506	24.236	13.276	20.756	11.576	10.176	10.906			41.676	15.386							
		american express	100.008	80.008						5.313											
		mastercard	100.008	55.568	29.236	18.486	50.006	14.396		10.216						7.323					
	debit	american express									14.295										
		mastercard	100.007	50.005	21.185		19.445	9.082		5.592	5.852					5.262					
		visa	100.007	34.295	14.065		6.152	8.592		5.812	7.162			25.005		15.525		5.412			
mobile	credit	american express				15.797		9.224		5.434	5.264										
		visa		53.339	24.267	29.017	29.637	18.507	11.117	15.197	7.584	50.007				8.974					
		discover	100.009			44.447		18.607									8.574				
		mastercard	100.009	81.829	20.597	20.637	16.677	25.067	12.507	16.267	13.727								10.437		
	debit	american express						9.093													
		mastercard			13.826	11.456	5.883	11.586	9.763	5.403									6.313		
		visa			8.373	10.536	6.253	11.486													

Q&A

Q: Why is there no mention of impossible travel in the data visualization?

A: While highly effective, the IEEE-CIS dataset has encrypted/hashed location attributes to protect user **PII (Personally Identifiable Information).** Without raw Latitude/Longitude or IP-to-City mapping, calculating geographic velocity was not feasible. I pivoted the focus to Device and Email Anonymity instead.

Q: Should we block all traffic from protonmail due to its high risk scores?

A: No. While protonmail is high risk indicator it is not a definitive proof of fraud instead my recommendation is to trigger **Enhanced Multi-factor authentication(MFA)** or a manual review of all protonmail users, especially if the user is on a desktop emulator

Recommendations

Strategic recommendations following the analysis:

- Implement a secure Multi-Factor Authentication(MFA) for all protonmail transactions
- Apply stricter velocity limits on Credit Card transactions originating from Desktop Emulators.
- Transition to behavior-based biometric monitoring to detect "non-human" interaction patterns e.g., Identifying **High-Velocity Attacks** where automated scripts attempt hundreds of transactions within a 5-10 second window.

Closing statement

Title: Key Takeaway

Fraud is no longer about single variables; it is about the intersection of anonymity and technology. By targeting the 'Triple Threat' of Protonmail, Desktop Emulation, and Credit Liquidity, we can proactively secure 90% of high-risk vectors.

Contact and Portfolio

Thank you for your Time

Robert Norris | Aspiring Data Analyst

SQL Queries links:

Emails, Devices, Cards, Overall risk score

Tableau Interactive dashboard links:

Emails, Devices, Cards, Overall risk score